Sun Li
Gender: Female   Objective: Senior Backend Engineer   Location: Beijing
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Education

2010.09 - 2014.06   Hunan University   Artificial Intelligence (B.S.)
Courses: Design Patterns, Software Engineering, Data Structures and Algorithms, Machine Learning, Operating Systems, Compilers

Summary

Seven years of backend experience focused on payment/transaction systems, distributed architecture, high concurrency and production incident response.

Work Experience

2017.07 - Present   NIO   Senior Backend Engineer
- Led a production incident review with root-cause analysis and drove an alert-tiering and on-call rotation mechanism.
- Investigated rising instance memory rss using GC logs, heap dump and arthas, found an unbounded cache and cut memory by 59%, recognized by stakeholders and peers.
- Fixed a log-collection failure caused by agent backpressure, restoring stability after tuning batch and buffer settings, supporting rapid business growth.
- Optimized a file-sort slow SQL on a hot API, added a composite index via gh-ost online DDL, reducing P99 from 1500ms to 420ms, significantly lowering maintenance cost.
- Built service observability covering API latency, error rate, JVM GC, thread pools and consumer backlog with Prometheus and Grafana, and key metrics improved steadily.

2014.07 - 2017.06   Lalamove   Backend Engineer
- Served 7000K requests per day while keeping core-path SLA above 99.9%, and related standards were adopted by the team.
- Tuned Redis caching for hot keys and cache breakdown with local cache and distributed locks, raising hit rate above 90%, with experience documented and shared internally.
- Rebuilt the config and template management service to support hot template reload without restart, cutting release time by 47%, supporting rapid business growth.
- Designed a Kafka-based async transaction pipeline with idempotency keys, retry queues and dead-letter handling, sustaining 8000 QPS at peak, recognized by stakeholders and peers.

Projects

High-Concurrency Flash-Sale & Inventory System (Java + Redis + RocketMQ)
- Pre-deducted inventory in Redis with async DB writes to absorb traffic spikes.
- Stayed stable at 8000 QPS in load tests with error rate below 0.1%.
- Built circuit breaking and fallback for downstream failures.
- Designed distributed locks and token-bucket rate limiting to prevent oversell.

Enterprise Auth & Gateway Platform (Spring Cloud Gateway + OAuth2)

- Integrated OAuth2 and RBAC with multi-tenant isolation.
- Shipped a unified SDK to lower integration cost across teams.
- Unified auth, rate limiting, gray routing and tracing as reusable platform capabilities, and key metric s improved steadily.
- Added circuit breaking to isolate faulty downstreams.

Distributed Payment & Settlement Platform (Spring Boot + MySQL + Kafka + Redis)

- Applied idempotency and eventual consistency to avoid duplicate charges on retry, reused across multipl e business lines.
- Recorded an audit trace id for every payment state transition for compliance.
- Implemented daily reconciliation across payment gateway, ledger and settlement files with auto-correcti on, recognized by stakeholders and peers.
- Used sharding and read-write splitting to keep single-table rows within tens of millions.

Highlights

- Drove a focused effort on Linux and shell basics, reducing related issues by about 47%
- Continuously optimized Redis caching and distributed locks, cutting key-path latency from 1500ms to 420 ms
- Owned core modules that steadily served 7000K daily visits with 99.9%+ availability
- Built a reusable methodology and docs around root-cause analysis and postmortems, adopted by several te ams
- Established metrics and reviews around Docker and containerization to keep improving delivery quality

Skills

Core skills: Kafka, Spring Boot, SQL Optimization, Redis caching and distributed locks, Kafka/RocketMQ me ssaging

Proficient in: Docker and containerization, JVM tuning and troubleshooting, MySQL, Microservices, Observa bility

Familiar with: JVM, MySQL indexing and slow-query tuning, Redis, Java, Distributed transactions and consi stency, High Concurrency

Self Evaluation

- Solid engineering fundamentals and strong troubleshooting skills, able to own a module end to end
- Enjoys retrospectives, cares about code quality and observability, and makes data-driven decisions